

Unit 6 – Inference for Categorical Data: Proportions

Notes 3 – Introduction to Significance Tests

Opening Activity

- Let's start off with a small activity: I need a volunteer!
- We are going to see how well _____ can play trashketball! They will make 10 throws of this balled up trash from a spot on the floor into the garbage can.
- _____, how many do you believe you can make out of 10? _____
- Class, do you believe he will make more or less than this? _____

We are going to test this claim!

Claim one: _____ Claim two: _____

These claims are called hypotheses and the test to determine which one we will go with is a significance test.

Claim one: _____ This claim is called our null hypothesis: _____

This can be a claim made by someone/something or a statement of "no difference"

Claim two: _____ This claim is called our alternative hypothesis, _____

This is the claim we are trying to find evidence for.

Let's have them shoot. What was their sample proportion? _____

Was it exactly the claimed proportion? Do we expect it to be?

Can they have gotten this sample proportion and their claim still be correct?

How can we decide if the difference is chance variation or if their original claim is incorrect?

One-Sided Hypothesis Test

- In significance testing, we set up two hypotheses: Example 1
 - The _____ hypothesis is an original claim, and is always the population proportion equal to a value
 - The _____ hypothesis is the claim we are trying to find evidence for Example 2
- In a one-sided hypothesis test, the inequalities are either _____ the value in our null hypothesis
- _____ is called our claimed proportion

Notice some similarities:

- The null hypothesis is always _____ a value
- The alternative is above or below that _____
- Our hypotheses are based on our population parameter _____

Two-Sided Hypotheses Test

- In a two-sided test, we are looking for evidence against the claim, but we are unsure about _____ Example
- Hypotheses are developed before a study is conducted, so after we gather $\hat{p}$ and we know which direction the difference is, we _____ and change our alternative hypothesis

A significance tests asks the question:

How likely is it to get a result like this just by chance, when the null hypothesis is true?

In our opening activity, if a student said they were a 80% trashketball shooter, it is possible for them to shoot 50% in a single game?

This is a conditional probability question: _____

If we find the conditional probability and it is small, this is good evidence _____

The probability of the event occurring given an original claim is called the _____

P-Value

- The _____, computed assuming H_0 is true, that the statistic $\hat{p}$ would take a value as extreme or more extreme than the one actually observed, in the direction specified by H_a
- We compute this probability by using our sample statistic $\hat{p}$ and the value in our null hypothesis, p_0
- We will be calculating z-scores to determine the difference between what we observed (_____) and what our claim is (_____)
- We will then calculate the probability we would get this value or more extreme (in the direction of the alternative) and make a decision:

if the probability is large enough, we could	if the probability is low enough, we could
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Significance Level

- How low is a low enough probability? This is determined by our _____
- This significance level is represented by the symbol _____, and researchers set it before the test. Choosing it is based on a few factors, which we will discuss later, but it is always given in a problem.

If our p-value is smaller than α , _____ we say that the

- result of the study is _____
- The probability of this event occurring by chance, if H_0 is true, is so low, we say that the difference is NOT due to sampling variability
- In this case, _____ and conclude that _____

If our p-value is larger than α , _____ we say that the

- result of the study is _____
- The difference between $\hat{p}$ and p_0 can be attributed to sampling variability
- In this case, _____ and conclude that _____

- This does NOT mean we “accept H_0 ”!! (NEVER SAY THIS). All we showed was that we do not have enough evidence against the claim. We did not prove the claim.

The usual significance levels are

- 1% ($\alpha =$ _____)
- 5% ($\alpha =$ _____)
- 10% ($\alpha =$ _____)

One-Sided Hypothesis Test		Two-Sided Hypothesis Test
$H_0: p = p_0$ $H_A: p < p_0$	$H_0: p = p_0$ $H_A: p > p_0$	$H_0: p = p_0$ $H_A: p \neq p_0$

Conducting a Significance Test

_____:

- Define p = in words
- Identify the hypotheses you are testing and write down your significance level

_____:

- You will be creating a sampling distribution, which requires you to check
 - Random (Random Sample or Random Assignment)
 - Independent (When sampling, check 10% condition)
 - Normal (Check large counts condition with $np_0 \geq 10$ and $n(1 - p_0) \geq 10$)

_____:

- Identify the inference you are performing: _____
- Calculate your test statistic:

Find your p-value by calculating the probability of getting a z-statistic this extreme or more extreme in the direction of the alternative hypothesis.

$H_A: p < p_0$	$H_A: p \neq p_0$	$H_A: p > p_0$

_____:

Because our p-value of <value> is <bigger/smaller> than our significance level of <level>, we <reject or fail to reject H_0 > . We <do/don't> have convincing evidence that the population proportion <context of what this means in the problem>